ΔContext-R: Budget-Aware Retrieval over an Online Topic Store for Long-Term Conversational Memory

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***Abstract*—Long-term chat assistants must answer from histories far longer than their context window. Recent segment-level memory stores a conversation as topical segments, denoises them by prompt compression and retrieves whole segments, but it needs a large-model call per session to segment, and whole segments exhaust small context budgets. This paper proposes ΔContext-R. The history is stored as topics by an online delayed-commitment segmenter that needs no language model and never revises a boundary. Retrieval ranks turns by fusing lexical, dense and topic evidence, then spends the budget on the user's own statements inside the most relevant topics. On LongMemEval-S, with one 2,048-token budget and one 3B reader, ΔContext-R raises evidence recall from 0.64 to 0.91 and answer F1 from 0.270 to 0.369 over segment-level memory, and at 4,096 tokens it beats strong turn-level hybrid retrieval (recall 0.969 vs. 0.912). Controlled ablations show that budget-aware reading of user turns drives the gain.**

***Index Terms*—Conversational memory, retrieval-augmented generation, topic segmentation, context budget, large language models, long-term dialogue.**

I. Introduction

Dialogue systems built on large language models (LLMs) are becoming long-lived assistants [1], [2], and an assistant that remembers earlier sessions needs a memory that grows without bound [3]. Placing the whole history in the prompt is not an answer: attention cost grows with input length [4], models use the middle of long inputs unreliably [5], and missing or irrelevant context causes hallucination [6]. Practical systems therefore retrieve, for each question, a few past passages under a token budget [7]–[9], summarise old turns [10], or manage memory with update and paging rules [11], [12].

How the history is cut into memory units matters. LongMemEval showed that long-context LLMs lose about 30% accuracy on chat histories of ∼115K tokens and that the granularity of stored values is a key design choice [13]. SeCom [14] went further: it segments every session into topics with an LLM, denoises each segment with LLMLingua-2 prompt compression [15], and retrieves whole segments, outperforming turn-level and session-level memory on two benchmarks. Topic structure is also known to help dialogue understanding [16], [17], and chunking choices change retrieval quality [18].

Problem statement. Segment-level memory has two costs. Building it takes one LLM call per session, and retrieving whole segments spends a small budget on a few long units, much of them assistant text. The question is how to store a growing chat as topics without an LLM, and how to retrieve from that store so that a fixed budget carries the most evidence.

Objective. To build a topic-indexed conversational memory with no LLM in the loop, and a retrieval rule that beats segment-level memory and strong turn-level retrieval at the same token budget and reader.

Contributions.

1) ΔContext-R, a budget-aware retrieval method over an online topic store. It fuses BM25, dense and topic ranks, reads the user's turns inside the most relevant topics, and reserves part of the budget for them.
2) A controlled re-implementation of SeCom and of turn-, exchange- and session-level memory, with every unit, key (raw or LLMLingua-2) and retriever evaluated under one budget, one reader and dev-only tuning.
3) On LongMemEval-S, ΔContext-R improves on SeCom by 0.27 evidence recall and 0.099 answer F1 (a 37% relative gain) and beats turn-level hybrid retrieval at 4,096 tokens, without any LLM call to build the memory.
4) An ablation showing what drives the gain: reading short user turns within the budget, not topic structure as such. The online topics match true sessions and beat fixed chunks, but topic-free budget rules perform equally well.

Organisation. Section II reviews related work, Section III presents ΔContext-R, Section IV reports results, Section V compares with existing works, and Section VI concludes.

II. Literature Review

Memory systems. MemoryBank [11] stores memories with summaries and a forgetting rule; MemGPT [12] lets the LLM page information between prompt and storage; recursive summarisation [10] rewrites a running memory; LongMemEval [13] decomposes sessions into rounds and expands keys with LLM-extracted facts. SeCom [14] is the closest

work: segment-level units, compression-based denoising with LLMLingua-2 [15], and BM25 or MPNet retrieval.

Topic segmentation. TextTiling [19] cuts at depth minima of a similarity curve and is the decoder of neural methods such as CSM [20]; DialSTART [21] and UMLF [17] learn topic-aware representations, and SuperDialseg [22] supplies supervised data. All segment complete dialogues offline. Semantic chunking [23] and the chunkers compared by Lavarec and Du [18] also need the whole text.

Retrieval. BM25 [24] and dense sentence encoders [25], [26] are the standard retrievers, and reciprocal-rank fusion (RRF) [27] combines them without calibration. Label-free semantic structure has also proven useful for named entities [28], relation extraction for knowledge graphs [29], aspect extraction with BERT-guided topic models [30] and topic-guided summarisation [31]. Table I summarises ten recent works.

Identified problem. Existing memories either retrieve fixed units that ignore the budget, or build topic units with an LLM and retrieve them whole. No prior work builds topics online without an LLM and decides, within a token budget, which turns of which topics to read, or tests what actually makes such retrieval work.

III. Proposed Work

ΔContext-R has two stages and four phases (Fig. 1). Storage (Phases 1–3) runs once per incoming turn; retrieval (Phase 4) runs once per question.

A. Phases 1–3: online topic storage

Each turn is prefixed with its role and encoded once by a sentence encoder f (all-MiniLM-L6-v2 [25]). Topics are found by ΔContext's delayed-commitment detector. With $\mu(i, j)$ the normalised mean of $e_i, \ldots, e_j$ and a the start of the open topic, gap g (between turns g and $g{+}1$) is scored once L further turns exist:

$$s_g = 1 - \mu\big(\max(a, g-c+1),\, g\big)^{\top} \mu\big(g+1,\, g+L\big). \quad (1)$$

Gap g is decided one turn later and becomes a boundary only if $s_g \geq s_{g-1}$, $s_g \geq s_{g+1}$ and $s_g \geq \tau$; a committed boundary is never revised, so the store is append-only. We use the setting tuned on LongMemEval dev histories for segmentation accuracy ($L{=}3$, $c{=}3$, $\tau{=}0.75$), fixed before any retrieval experiment. Each topic keeps its turns, the role and token cost of every turn, its text for BM25, and its centroid. No LLM is called at any point.

B. Phase 4: budget-aware topic retrieval

Algorithm 1 gives the procedure. For a question q, each turn i receives a hybrid score s_i, the RRF [27] of its BM25 rank [24] and its MiniLM cosine rank; each topic j receives t_j, the RRF of the BM25 rank of its text and the cosine rank of its centroid. With $r(\cdot)$ the rank (0 = best) and $K{=}60$, the fused score is

$$f_i = \frac{1}{K+1+r(s)_i} + \frac{w}{K+1+r(t)_{\tau(i)}} + \frac{\beta}{K+1}\,\mathbf{1}\big[\mathrm{user}(i) \wedge r(t)_{\tau(i)} < k\big], \quad (2)$$

where $\tau(i)$ is the topic of turn i. The first term ranks turns, the second prefers turns in relevant topics, and the third reads what the user said in the k most relevant topics. Turns are packed greedily by f into $(1-\rho)B$ tokens; the reserved ρB then takes the remaining user turns of every topic already read from, and any leftover budget takes other turns by f. The context is shown in chronological order with session dates.

Algorithm 1 ΔContext-R retrieval (per question)

Input: topic store; question q; budget B; w, β, k, ρ
Output: context turns in chronological order

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s_i ← RRF(BM25 rank, MiniLM rank) for every turn i
t_j ← RRF(BM25 rank of topic text, centroid rank) for every topic j
f_i ← Eq. (2); order turns by f
C ← greedy fill of (1−ρ)B tokens in f order (skip turns that do not fit)
T ← {τ(i) : i ∈ C}                          ▷ topics already read from
for user turns i ∉ C with τ(i) ∈ T, in f order do
    add i to C if it fits in B
add remaining turns in f order while they fit in B
return C sorted by time
```

Why read the user. In a personal-assistant memory the facts asked about are mostly what the user said, and user turns are short while assistant replies are long. A ranking that ignores cost spends the budget on a few long replies. Section V shows that two thirds of the evidence a turn ranker misses lies inside topics it already read from, which is what the reserve recovers.

Parameters and cost. $w{=}0.25$, $\beta{=}0.5$, $k{=}3$ and $\rho{=}0.25$ were chosen on dev questions only. Building the memory costs one MiniLM encoding and an $O((c+L)d)$ update per turn; answering costs one BM25 query and one cosine pass, with no LLM call.

IV. Results

A. Experimental setup

All experiments ran on one laptop (RTX 4060, 8 GB) in ContextBench, a harness in which only the memory policy varies: every method shares the token counter, the prompt with session dates, and the reader, Qwen2.5-3B-Instruct [32] (bf16, greedy). Budgets are 2,048 and 4,096 context tokens. Every method's configuration was chosen on the dev half by evidence recall at 2,048 tokens, then scored once on the test half. Metrics: evidence recall (share of labelled evidence turns that reach the context), all-evidence rate (questions with every evidence turn retrieved), token F1 and exact match (EM) of answers. Differences use a paired bootstrap (10,000 resamples) with Holm correction [33].

TABLE I
TEN RECENT RELATED WORKS AND THEIR DRAWBACK FOR A BUDGET-LIMITED CONVERSATIONAL MEMORY.

Work	Year	Algorithm	Advantage	Drawback for our setting
MemoryBank [11]	2024	Memory store, summaries, forgetting curve, dense retrieval	Persistent, user-adaptive	Flat units; no budget-aware selection
MemGPT [12]	2023	LLM-managed paging between prompt and storage	Unbounded virtual context	An LLM call per memory operation
Recursive summaries [10]	2025	LLM rewrites memory from old memory and new turns	Compact memory	Lossy; verbatim evidence is lost
LongMemEval [13]	2025	Round-level values, fact-augmented keys, time-aware queries	Systematic memory study	Key expansion needs an LLM per round
SeCom [14]	2025	LLM segmentation, LLMLingua-2 denoising, segment retrieval	Beats turn and session memory	LLM call per session; whole segments fill small budgets
LLMLingua-2 [15]	2024	Token classification keeps informative tokens	Fast, faithful compression	A 560M-parameter model over all stored text
UMLF [17]	2025	Joint discourse parsing and topic segmentation	Best unsupervised segmentation	Offline; several large encoders
DialSTART [21]	2023	Topic-aware contrastive utterance learning	Strong unsupervised accuracy	Needs in-domain training; offline
Topic-aware comprehension [16]	2024	Unsupervised topic blocks + dual-attention matching	Topics aid response selection	Segments finished passages; no retrieval budget
Chunking for RAG [18]	2026	Fixed, recursive, semantic, LLM chunking compared	Chunking changes RAG quality	Documents with structure that chats lack

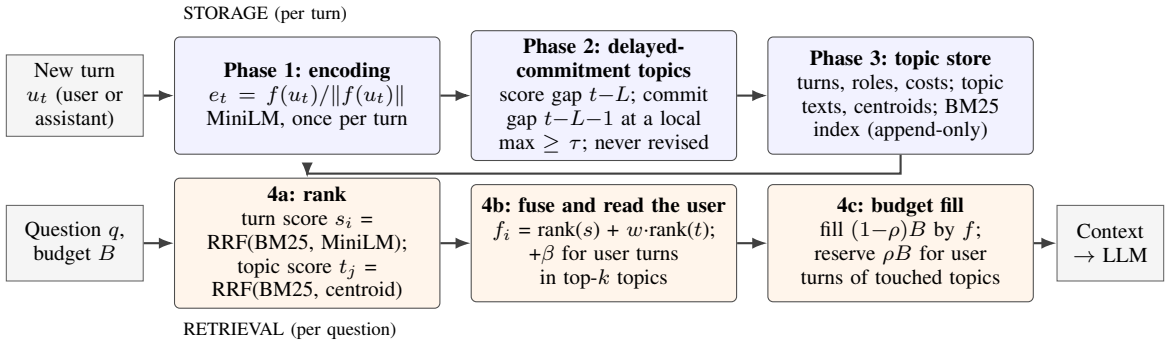


Fig. 1. ΔContext-R. Storage builds a topic store online with no language model; retrieval ranks turns with lexical, dense and topic evidence and spends the budget on the user's statements inside the most relevant topics.

B. Dataset

LongMemEval-S [13] gives each of 500 questions its own history of user–assistant sessions with turn-level evidence labels. A seeded random split gives 250 dev and 250 test histories; 233 and 237 of their questions are answerable and used here. Table II lists the memory each method stores. Topic quality was measured in prior work on LongMemEval and three dialogue benchmarks [22], [34], [35]; Table III recalls the numbers relevant to storage.

TABLE II
LONGMEMEVAL-S TEST HISTORIES: MEMORY UNITS PER HISTORY AND MEAN QWEN TOKENS PER UNIT.

Unit	Units / history	Tokens / unit
Turn (all)	491.9	225
user turn	243.9	**69**
assistant turn	247.9	379
Exchange (user turn + replies)	248.0	447
ΔContext topic (online)	47.8	2,322
SeCom unit (topic cut at sessions)	51.7	2,146
Session	47.6	2,331

C. Retrieval and answer quality

Fig. 2 and Table IV give the test results. ΔContext-R reaches evidence recall 0.908 at 2,048 tokens and 0.969 at 4,096, and retrieves every evidence turn for 85.7% and 94.1% of questions. It is the most accurate system at 4,096 tokens. At 2,048 tokens it fits 25.3 turns into the budget on average, where turn-level retrieval fits 11.6 and SeCom 11.2, because it spends the budget on short user statements (69 tokens on average, against 379 for an assistant reply; Table II). Answer F1 follows recall (Fig. 3): 0.369 at 2,048 tokens, the highest of all systems, against 0.270 for SeCom. Doubling the budget raises recall for every system but lowers F1 for all of them: the 3B reader handles 4K-token contexts worse, a long-context effect consistent with [5]. ΔContext-R degrades least (0.310, against 0.258 for turn-level hybrid retrieval and 0.260 for SeCom), so at 4,096 tokens its F1 lead over turn-level retrieval becomes significant ($+0.052$, $[+0.015, +0.090]$,

TABLE III
STORAGE QUALITY: TOPIC BOUNDARIES AGAINST SESSION BOUNDARIES ON 250 LONGMEMEVAL-S TEST HISTORIES (PARAMETERS TUNED ON DEV). REVISIONS: EXISTING BOUNDARIES CHANGED PER 100 NEW TURNS WHEN THE HISTORY IS REPLAYED ONE TURN AT A TIME.

Segmenter	Online	Pk↓	$F1_0$ ↑	Revisions	LLM calls
ΔContext (ours)	yes	**0.058**	**0.914**	**0**	0
TextTiling [19]	no	0.069	0.894	17.0	0
Semantic chunking [23]	no	0.141	0.756	17.6	0
SeCom (GPT-4) [14]	no	–	–	–	1 / session

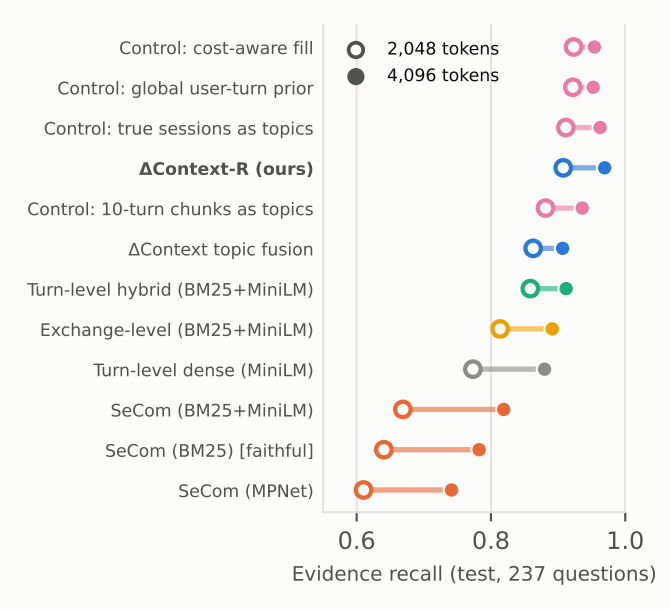


Fig. 2. Evidence recall at 2,048 (open) and 4,096 (filled) context tokens on 237 test questions. Blue: ΔContext; orange: SeCom; pink: topic-free and grouping controls (Section V).

p=0.02). With this reader the best budget for every system is the smaller one.

D. Question types

Fig. 4 breaks recall down by question type. The largest gains are on multi-session questions, whose evidence is spread over several sessions (0.869 vs. 0.732 for turn-level hybrid retrieval and 0.387 for SeCom), and on preference questions (1.000 vs. 0.678 and 0.567). Reading the user costs recall where the evidence is an assistant reply (0.815 vs. 0.889 at 2,048 tokens), the price of the user prior; at 4,096 tokens both reach 1.000.

V. COMPARISONS

A. Compared works

SeCom [14] is re-implemented from its code: memory units are topic segments, keys are their LLMLingua-2 compressions [15] (xlm-roberta-large, rate 0.75, the turns of a session compressed jointly), retrieval returns whole units ranked by BM25 (its best retriever) or MPNet [26], and units fill the budget in rank order. SeCom's GPT-4 segmentation needs one API call for each of LongMemEval-S's 19,195 sessions, so we

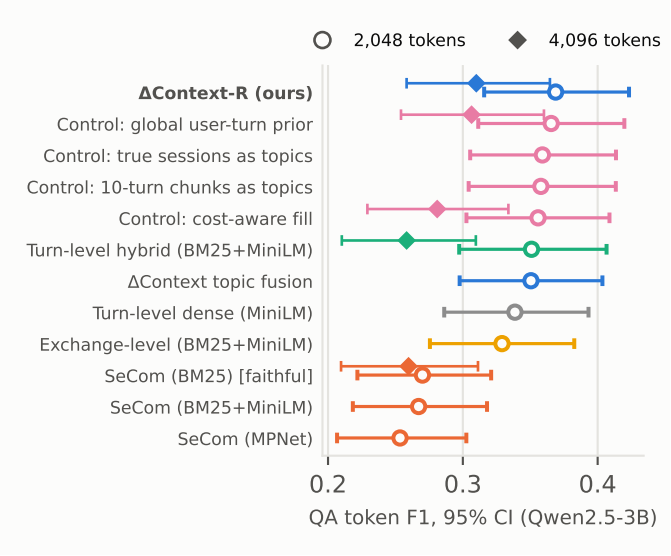


Fig. 3. Answer token F1 with 95% bootstrap intervals (Qwen2.5-3B reader, test): circles at 2,048 tokens, diamonds at 4,096 tokens (run for the main systems). Every system answers worse with the larger budget.

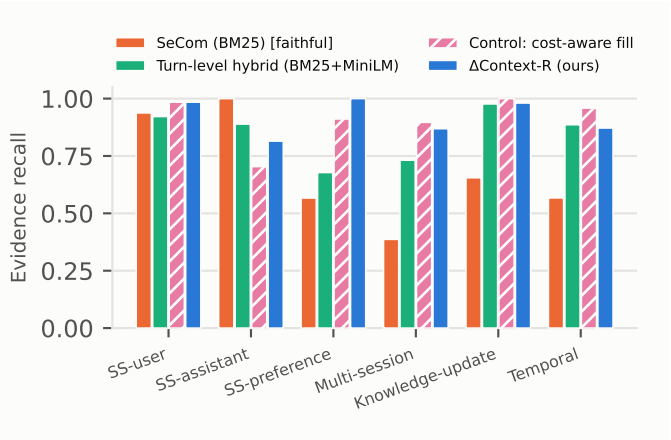


Fig. 4. Evidence recall by question type at 2,048 tokens (test).

give it ΔContext's topics cut at session starts, the closest LLM-free equivalent. We also give it our stronger BM25+MiniLM retriever. **LongMemEval's round-level memory** [13] is the exchange-level baseline (a user turn and its replies). **Turn-level retrieval** uses dense MiniLM or BM25+MiniLM fusion [27], the strongest baseline here.

B. ΔContext-R against SeCom

Against faithful SeCom, ΔContext-R improves recall by 0.267 (95% CI $[+0.220, +0.314]$) at 2,048 tokens and 0.187 ($[+0.149, +0.228]$) at 4,096, and F1 by 0.099 ($[+0.050, +0.148]$) at 2,048 tokens, all $p < 10^{-3}$ (at 4,096 tokens the F1 gap is $+0.050$, p=0.07 after correction), while building its memory with no LLM call and no compression pass (SeCom's LLMLingua-2 pass alone took about two GPU hours for the 47M tokens of LongMemEval-S, before any segmentation call). Fig. 5 shows why segment-level memory

TABLE IV

COMPARISON ON 237 LONGMEMEVAL-S TEST QUESTIONS. RECALL: EVIDENCE RECALL; ALL: SHARE OF QUESTIONS WITH ALL EVIDENCE RETRIEVED; TURNS: MEAN TURNS IN THE 2,048-TOKEN CONTEXT. DIFFERENCES TO ΔCONTEXT-R IN RECALL, F1 AND EM: $^{***}p < 10^{-3}$, $^{*}p < 0.05$ (PAIRED BOOTSTRAP, HOLM); OTHERWISE NOT SIGNIFICANT; –: NOT RUN.

		2,048 tokens				4,096 tokens				
Memory	Unit / keys / retriever	Recall	All	F1	EM	Recall	All	F1	Turns	LLM calls
SeCom [14] (faithful)	segment / LLMLingua-2 / BM25	0.641***	0.443	0.270***	0.173*	0.782***	0.667	0.260	11.2	1 / session
SeCom (MPNet)	segment / LLMLingua-2 / MPNet	0.610***	0.426	0.253***	0.160*	0.741***	0.608	–	11.6	1 / session
SeCom (+ our retriever)	segment / LLMLingua-2 / BM25+MiniLM	0.669***	0.494	0.267***	0.169*	0.819***	0.726	–	11.1	1 / session
Round-level [13]	exchange / raw / BM25+MiniLM	0.814***	0.700	0.329	0.224	0.891***	0.814	–	11.4	0
Turn-level dense	turn / raw / MiniLM	0.773***	0.658	0.339	0.232	0.880***	0.797	–	12.6	0
Turn-level hybrid	turn / raw / BM25+MiniLM	0.859	0.785	0.351	0.253	0.912***	0.848	0.258*	11.6	0
ΔContext topic fusion	turn+topic / raw / BM25+MiniLM	0.863	0.789	0.351	0.245	0.907***	0.840	–	11.5	0
ΔContext-R (ours)	turn+topic / raw / BM25+MiniLM	**0.908**	**0.857**	**0.369**	**0.253**	**0.969**	**0.941**	**0.310**	25.3	**0**

loses here. With every retriever, recall falls as units grow: turns and exchanges reach 0.66–0.82, topics and sessions only 0.48–0.60, because a LongMemEval session (∼2.4K tokens) barely fits in the budget and most of it is assistant text. Finer LLM segments would lie between exchanges and sessions in size, and even exchanges with the best retriever reach only 0.814 on test, 0.094 below ΔContext-R. LLMLingua-2 denoising does what SeCom reports for dense retrievers (MPNet on turns: 0.67→0.75; on topics: 0.48→0.54), but it does not close the granularity gap.

C. ΔContext-R against turn-level retrieval

Turn-level hybrid retrieval is a much stronger baseline than SeCom on LongMemEval. ΔContext-R beats it by 0.057 recall at 4,096 tokens ($[+0.031, +0.086]$, $p < 10^{-3}$) and by 0.049 at 2,048 ($[+0.010, +0.087]$; not significant after Holm correction). In answers it ties at 2,048 tokens (F1 0.369 vs. 0.351, n.s.) and wins at 4,096 (0.310 vs. 0.258, p=0.02). Topic fusion alone (without reading the user) gives no gain (0.863 vs. 0.859).

D. What drives the gain

Table V replaces one ingredient at a time, each control tuned on dev with its own grid. Using true sessions instead of ΔContext's online topics gives the same result, and 10-turn chunks are worse (significant at 4,096 tokens), so the online topics are as good a grouping as the ground truth. Two topic-free rules, a global bonus for user turns and cost-aware filling (rank score divided by token cost$^{0.25}$), match ΔContext-R: at 2,048 tokens they are 0.014–0.015 higher and at 4,096 0.015–0.017 lower, none significant. The gain therefore comes from budget-aware reading of user statements; the topic structure contributes on preference questions (Fig. 4), where topic-based variants reach 1.000 against 0.778–0.911 for topic-free rules, but not overall. A diagnostic on dev explains the design: of the evidence turns that topic fusion misses at 2,048 tokens, 67% lie inside topics it already read from (78% for multi-session questions), so the missing evidence is a user statement that was ranked low, not an unseen topic. Penalising repeated topics to spread the budget made recall worse at every strength (0.80–0.60 vs. 0.83 on dev).

TABLE V

ABLATION ON TEST: RECALL OF ΔCONTEXT-R MINUS EACH CONTROL (95% CI). $^{***}p < 10^{-3}$ (HOLM); OTHERS N.S.

Control (tuned on dev)	2,048 tokens	4,096 tokens
True sessions as topics	−.004 [−.029, +.021]	+.007 [−.006, +.021]
10-turn chunks as topics	+.026 [−.005, +.059]	+.033*** [+.016, +.054]
Global user prior (no topics)	−.014 [−.050, +.020]	+.017 [−.006, +.042]
Cost-aware fill (no topics/roles)	−.015 [−.053, +.023]	+.015 [−.008, +.041]
Topic fusion (no user reading)	+.045 [+.007, +.082]	+.063*** [+.035, +.091]

E. Limitations

Results come from one benchmark, whose sessions are mostly single-topic and whose evidence is mostly user statements; the user prior helps there and costs recall where evidence is an assistant reply. The reader is a 3B model scored by token F1, not an LLM judge, and it answers worse with 4K than with 2K tokens of context, so recall gains at the larger budget do not all reach the answer. SeCom's GPT-4 segmentation was replaced by an LLM-free equivalent, and the exchange-level baseline bounds what finer segments could reach. Dev selection over many configurations made dev gains optimistic, which is why every claim above rests on the single test pass.

VI. CONCLUSION

This paper presented ΔContext-R, a conversational memory that stores a growing chat as topics found online, with no language model, and retrieves by reading the user's statements inside the most relevant topics under a fixed token budget. Against segment-level memory with compression-based denoising, it raised evidence recall from 0.64 to 0.91 and answer F1 by 37% at the same budget and reader, and at the larger budget it beat strong turn-level hybrid retrieval. Its novelty is a retrieval rule shaped by the budget: rank turns with lexical, dense and topic evidence, then spend the budget on short user statements in the topics already judged relevant. Controlled ablations showed that this budget-aware reading, rather than topic structure itself, drives the gain, while the online topics proved as good a grouping as the true sessions. Future work will test the rule on benchmarks whose sessions mix several

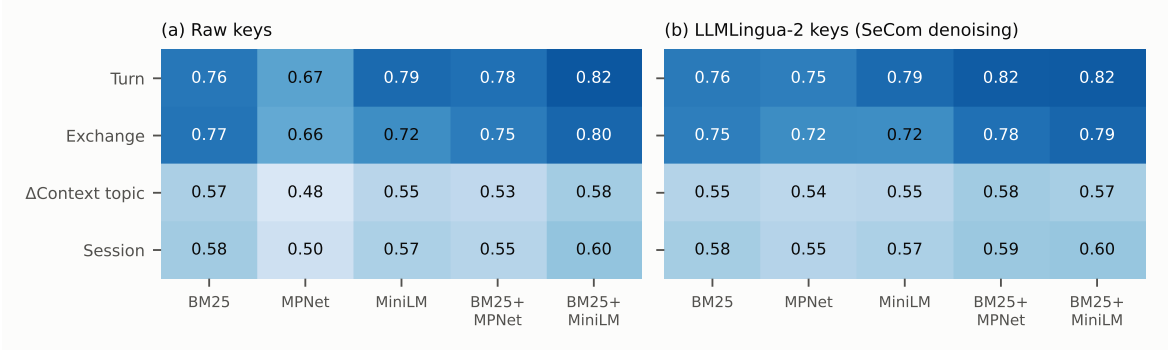


Fig. 5. Dev evidence recall at 2,048 tokens when whole units are retrieved, by memory unit and retriever, with raw keys (a) and SeCom's LLMLingua-2 keys (b). Larger units lose with every retriever; compression mainly helps MPNet.

topics, and learn the user prior and budget split from the question instead of fixing them.

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